

Generic Degree Three in $JC(3)$: Collision Geometry and Galois Group S_3

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Abstract

We study possible counterexamples to the complex three-dimensional Jacobian conjecture. For a polynomial map $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, let R_F be the scheme of pairs (x, y) satisfying $F(x) = F(y)$, and let C_F be its coordinate ring. The pairs (x, x) form a closed diagonal in R_F , whose defining ideal in C_F we denote by $\text{Obs}(F)$. The coordinate functions of F generate

$$B = \mathbb{C}[F_1, \dots, F_n] \subseteq A = \mathbb{C}[x_1, \dots, x_n],$$

and hence a function-field extension $K = \text{Frac}(B) \subseteq L = \text{Frac}(A)$. We prove that every three-dimensional Keller map with $[L : K] = 3$ is a counterexample to $JC(3)$.

For every such map, the generic collision algebra decomposes as

$$K \otimes_B C_F \simeq L \times N,$$

where N is the normal closure of L/K . Consequently, the normal closure of the function-field extension attached to every generic-degree-three counterexample to $JC(3)$ has Galois group S_3 .

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1 Introduction

The complex Jacobian conjecture $JC(n)$ asks whether every polynomial map

$$F = (F_1, \dots, F_n): \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^\times,$$

has a polynomial inverse. Such an F is called a Keller map. The Jacobian condition makes F an étale morphism; the conjecture asks whether this local condition forces global invertibility.

The equations

$$F_i(x) - F_i(y) = 0 \quad (1 \leq i \leq n)$$

define a closed collision scheme

$$R_F \subseteq \mathbb{A}_{\mathbb{C}}^n \times \mathbb{A}_{\mathbb{C}}^n.$$

Its complex points are the pairs (x, y) with $F(x) = F(y)$; equivalently, R_F is the fiber square of F . It contains the diagonal Δ , consisting of the pairs (x, x) . If C_F is the coordinate ring of R_F , restriction to the diagonal gives

$$\bar{\mu}_F: C_F \longrightarrow A, \quad A = \mathbb{C}[x_1, \dots, x_n],$$

and we write $\text{Obs}(F) = \ker(\bar{\mu}_F)$. By Ax–Grothendieck,

$$\text{Obs}(F) = 0 \iff R_F = \Delta \iff F \text{ is a polynomial automorphism.}$$

A possible counterexample must therefore remain globally noninvertible despite being étale. We organize this failure by generic degree. On coordinate rings, F induces

$$F^*: \mathbb{C}[t_1, \dots, t_n] \longrightarrow A, \quad t_i \longmapsto F_i.$$

Let

$$B = \text{im}(F^*) = \mathbb{C}[F_1, \dots, F_n] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

For a Keller map, the F_i are algebraically independent, so F^* identifies the target coordinate ring with B . Thus $K = \mathbb{C}(F_1, \dots, F_n) \subseteq L = \mathbb{C}(x_1, \dots, x_n)$ is the induced finite separable extension of function fields, and $[L : K]$ is the generic degree. Classical Galois rigidity makes such a map an automorphism whenever L/K is normal. Since extensions of degree one and separable extensions of degree two are normal, Galois rigidity excludes both degrees. Generic degree three is therefore the first possible degree of a counterexample.

We henceforth study polynomial maps $F: \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ with generic degree $[L : K] = 3$. Here $JC(3)$ denotes the conjecture in three variables, while $[L : K] = 3$ is the separate generic-degree condition. Since polynomial automorphisms have generic degree one, every map with $[L : K] = 3$ is already nonautomorphic. In this generic-degree-three case,

$$F \text{ is a counterexample to } JC(3) \iff F \text{ is Keller.}$$

The residual decomposition itself is field-theoretic and does not use the Keller condition. For a map of generic degree three, the generic collision algebra has a diagonal factor L and a quadratic residual factor D_α :

$$K \otimes_B C_F \simeq L \times D_\alpha.$$

The quadratic residual factor is split in the normal branch and is a field in the nonnormal branch. In the latter case it is L -isomorphic to the normal closure N . Galois rigidity excludes the normal branch for Keller maps, so every generic-degree-three counterexample to $JC(3)$ has

$$K \otimes_B C_F \simeq L \times N, \quad \text{Gal}(N/K) \simeq S_3.$$

Section 2 develops the dimension-independent collision obstruction, generic base change, marked-root decomposition, and automorphism criterion. Section 3 specializes these constructions to cubic extensions. Section 4 isolates the normalization and boundary questions leading to the planar sequel. The closing low-degree corollary records the precise handoff to that paper.

2 General construction

For a fixed dimension n , where can $JC(n)$ fail? To answer this, we attach to an arbitrary polynomial map the scheme of pairs with the same image and the ideal defining its diagonal. A Keller map is a counterexample to $JC(n)$ precisely when this ideal is nonzero. The construction itself is dimension-independent and does not require the Keller condition.

2.1 The two ideals and their obstruction

Throughout the paper, the base field is $\mathbb{C}$. Let

$$F = (F_1, \dots, F_n): X = \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n$$

be a polynomial map, and let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The map F is Keller if

$$\det JF \in \mathbb{C}^\times.$$

Define

$$I_R(F) = (F_i(x) - F_i(y))_{i=1}^n, \quad I_\Delta = (x_i - y_i)_{i=1}^n, \quad C_F := S/I_R(F).$$

For a fixed map F , we abbreviate $I_R(F)$ to I_R .

Proposition 2.1 (Canonical containment). *For every polynomial map F ,*

$$I_R(F) \subseteq I_\Delta.$$

Proof. For $H \in \mathbb{C}[x_1, \dots, x_n]$, the difference $H(x) - H(y)$ belongs to $(x_1 - y_1, \dots, x_n - y_n)$. This follows by induction on H , or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate F_i . $\square$

Let $q_F: S \rightarrow C_F$ be the quotient map. The containment $I_R \subseteq I_\Delta$ allows us to define the obstruction ideal

$$\text{Obs}(F) := q_F(I_\Delta) = I_\Delta/I_R \subseteq C_F.$$

Let $A := \mathbb{C}[x_1, \dots, x_n]$. Diagonal evaluation $S \rightarrow A$, given by $s(x, y) \mapsto s(x, x)$, is surjective with kernel I_Δ . Since $I_R \subseteq I_\Delta$, it factors canonically as

$$S \xrightarrow{q_F} C_F \xrightarrow{\bar{\mu}_F} A, \quad [s]_{I_R} \mapsto s(x, x).$$

Proposition 2.2 (The diagonal kernel). *As ideals of C_F ,*

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta/I_R.$$

Equivalently,

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof. The construction of $\bar{\mu}_F$ identifies its kernel with $I_\Delta/I_R = \text{Obs}(F)$. Since $I_R \subseteq I_\Delta$,

$$I_\Delta/I_R = 0 \iff I_R = I_\Delta.$$

□

2.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let $X = \text{Spec } A$, let $Y = \text{Spec } B$, and let

$$f_F: X \longrightarrow Y$$

be the morphism induced by $B \hookrightarrow A$. The homomorphisms $\iota_x, \iota_y: A \rightrightarrows C_F$, defined by $\iota_x(a) = [a(x)]$ and $\iota_y(a) = [a(y)]$, agree on B because $[F_i(x)] = [F_i(y)]$ in C_F . The equality $\iota_x|_B = \iota_y|_B$ defines the B -algebra structure on C_F .

Theorem 2.3 (Fiber-product interpretation). *There is a canonical B -algebra equivalence and its dual affine-scheme equivalence*

$$C_F \cong A \otimes_B A, \quad R_F := \text{Spec } C_F \cong X \times_Y X.$$

Under the first equivalence, $\bar{\mu}_F$ is tensor multiplication

$$A \otimes_B A \longrightarrow A, \quad a \otimes a' \longmapsto aa'.$$

Its dual morphism is the diagonal

$$\Delta_{X/Y}: X \longrightarrow X \times_Y X.$$

Proof. A map $C_F \rightarrow T$ to a commutative $\mathbb{C}$ -algebra T is equivalent to a pair of maps $f, g: A \rightrightarrows T$ satisfying

$$f(F_i) = g(F_i) \quad (1 \leq i \leq n).$$

Since the F_i generate B , this is precisely the universal property of $A \otimes_B A$. Applying Spec to this equivalence and using

$$\text{Spec}(A \otimes_B A) \cong \text{Spec } A \times_{\text{Spec } B} \text{Spec } A$$

gives the asserted fiber-product equivalence. The statements about multiplication and the diagonal follow on pure tensors. □

On complex points,

$$R_F(\mathbb{C}) = \{(u, v) : F(u) = F(v)\}.$$

Inside $X \times X = \text{Spec } S$, the ideals I_R and I_Δ define $X \times_Y X$ and its diagonal. Thus

$$I_R = I_\Delta \iff X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes. This is the geometric form of obstruction vanishing.

2.3 Generic fibers and marked-root decomposition

Let F be a polynomial map, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

Proposition 2.4 (Keller generic geometry). *If F is Keller, then its coordinate functions are algebraically independent, the morphism*

$$f_F: \text{Spec } A \longrightarrow \text{Spec } B$$

is étale and quasi-finite, and L/K is a finite separable extension. Consequently, every Keller map, and in particular every complex Jacobian-conjecture counterexample, has a finite separable generic field extension.

Proof. The identity

$$dF_1 \wedge \dots \wedge dF_n = \det(JF) dx_1 \wedge \dots \wedge dx_n \neq 0$$

gives algebraic independence, so $B \cong \mathbb{C}[u_1, \dots, u_n]$. As a B -algebra, A has the presentation

$$A \cong \mathbb{C}[u_1, \dots, u_n, X_1, \dots, X_n] / (F_i(X) - u_i)_{i=1}^n.$$

Its relative Jacobian determinant is the unit $\det(JF)$. The Jacobian criterion therefore makes f_F étale, hence quasi-finite. Since $B \subseteq A$, the morphism is dominant, so quasi-finiteness gives $[L : K] < \infty$. Its generic point is étale, and therefore L/K is separable. $\square$

For the remainder of this subsection, assume that F is dominant and generically finite. Let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a.$$

Theorem 2.5 (Generic collision bridge). *The map θ is an isomorphism. Consequently, there is a K -algebra isomorphism*

$$K \otimes_B C_F \cong L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1 \otimes \bar{\mu}_F} & K \otimes_B A \\ \sim \downarrow & & \downarrow \theta \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

Proof. Let $U = B \setminus \{0\}$. Since $K = U^{-1}B$, localization gives

$$K \otimes_B A \cong U^{-1}A \subseteq L,$$

and under this identification θ is the inclusion. Since L/K is finite, each x_i is algebraic over K ; hence

$$U^{-1}A = K[x_1, \dots, x_n]$$

is a finite-dimensional domain over K , and therefore a field. It contains A , so it contains $\text{Frac}(A) = L$. Thus $U^{-1}A = L$, and θ is an isomorphism.

Using Theorem 2.3 and standard tensor base change gives

$$K \otimes_B C_F \cong K \otimes_B (A \otimes_B A) \cong (K \otimes_B A) \otimes_K (K \otimes_B A) \xrightarrow{\theta \otimes \theta} L \otimes_K L.$$

Under this composite, both routes around the square send $\lambda \otimes (a \otimes a')$ to $\lambda a a'$. Hence the square commutes. $\square$

Suppose now that L/K is separable and has a power basis $L = K(\alpha)$. Let $m_\alpha(T) \in K[T]$ be the minimal polynomial of α , and let $h_\alpha(T) \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Theorem 2.6 (Marked-root tensor decomposition). *Give $L \otimes_K L$ its L -algebra structure through the left tensor factor. There is an L -algebra equivalence*

$$L \otimes_K L \cong L \times L[T]/(h_\alpha),$$

under which tensor multiplication $L \otimes_K L \rightarrow L$ is projection onto the first factor.

Proof. Via the power basis, scalar extension identifies

$$L \otimes_K L \xrightarrow{\sim} L[T]/(m_\alpha), \quad \ell \otimes p(\alpha) \mapsto [\ell p(T)],$$

where m_α is viewed in $L[T]$. Separability gives $h_\alpha(\alpha) = m'_\alpha(\alpha) \neq 0$, so $T - \alpha$ and h_α are coprime. The Chinese remainder theorem therefore gives

$$L[T]/(m_\alpha) \cong L[T]/(T - \alpha) \times L[T]/(h_\alpha) \cong L \times L[T]/(h_\alpha).$$

Under the first identification, tensor multiplication is evaluation at $T = \alpha$; under the Chinese remainder isomorphism this is projection onto $L[T]/(T - \alpha) \cong L$, the first factor. $\square$

Together with Theorem 2.5, this gives

$$K \otimes_B C_F \cong L \times L[T]/(h_\alpha),$$

where the first projection is the generic diagonal.

2.4 Automorphism detection

Theorem 2.7 (Automorphism criterion). *For an arbitrary polynomial map $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, the following are equivalent:*

- (1) F is a polynomial automorphism;
- (2) F is injective on $\mathbb{A}^n(\mathbb{C})$;
- (3) $I_R(F) = I_\Delta$;
- (4) $\text{Obs}(F) = 0$.

Proof. Suppose first that F has polynomial inverse $G = (G_1, \dots, G_n)$. Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every i , so $I_\Delta \subseteq I_R$. Proposition 2.1 gives equality. Thus (1) implies (3).

Suppose $I_R = I_\Delta$, and let $\text{ev}_{(a,b)}: S \rightarrow \mathbb{C}$ be evaluation at (a, b) . If $F(a) = F(b)$, then

$$I_R \subseteq \ker(\text{ev}_{(a,b)}).$$

Therefore $x_i - y_i \in I_\Delta = I_R$ gives $a_i - b_i = 0$ for every i . Thus $a = b$, proving (3) $\Rightarrow$ (2). The Ax–Grothendieck theorem [Ax69] gives (2) $\Rightarrow$ (1). Finally, Proposition 2.2 gives (3) $\iff$ (4). $\square$

Applied to Keller maps, Theorem 2.7 says that $JC(n)$ is equivalent to universal vanishing of $\text{Obs}(F)$, or equivalently to $I_R(F) = I_\Delta$.

We will also use the classical Galois case of the conjecture. It is the step that rules out the normal cubic branch under the Keller hypothesis.

Theorem 2.8 (Classical Galois rigidity). *Let*

$$F: \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^\times,$$

and put

$$K = \mathbb{C}(F_1, \dots, F_n), \quad L = \mathbb{C}(x_1, \dots, x_n).$$

If L/K is normal, then F is a polynomial automorphism and $L = K$.

Proof. The Keller condition makes L/K finite and separable. Thus normality makes it Galois, and the conclusion is the classical Galois case of the Jacobian conjecture due to Campbell, with algebraic treatments by Razar and Wright [Cam73, Raz79, Wri81]. $\square$

Lean correspondence. `CollisionIdeals/General/Collision/Diagonal.lean`,
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`CollisionIdeals/General/GenericFiber/MarkedRootDecomposition.lean`,
`CollisionIdeals/General/Keller/Interfaces.lean`,
`CollisionIdeals/General/Automorphism/Equivalences.lean`, and
`CollisionIdeals/General/Automorphism/GaloisRigidity.lean`. The polynomial fiber-product module uses the abstract target polynomial ring; the equivalence over the actual image algebra $B = \mathbb{C}[F_1, \dots, F_n]$ and its generic base change are owned by the generic-function-field module. The Lean form of Theorem 2.5 takes surjectivity of θ as a parameter; the proof above derives it from generic finiteness. The predicates `KellerEtaleBridge` and `KellerFlatBridge` expose parts of Proposition 2.4 as theorem targets; Lean does not yet derive them, generic finiteness, or separability from `IsKeller`. Ax–Grothendieck also remains an explicit parameter of the Lean automorphism equivalence.

3 The cubic S_3 obstruction in dimension three

We now specialize the dimension-independent constructions of Section 2 to generic degree three. After the diagonal is exhibited as the first factor, the complementary generic collision algebra has degree two over L . For a map of generic degree three it is either split, corresponding to a

normal cubic extension, or connected, corresponding to a nonnormal cubic extension. Degree three already rules out polynomial automorphy, so within this locus the counterexamples to $J\mathcal{C}(3)$ are precisely the Keller maps. Classical Galois rigidity rules out the split branch on that counterexample locus. Its residual factor is therefore the normal closure, whose Galois group is S_3 .

Let $S_3 = \text{Perm}(\{1, 2, 3\})$ be the symmetric group on three letters.

3.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3): \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \quad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

Assume that F has generic degree three:

$$[L : K] = 3.$$

Because K has characteristic zero, L/K is separable. Choose a primitive element α [The26, Tag 030N], so that $L = K(\alpha)$ with power basis $1, \alpha, \alpha^2$. If $m_\alpha \in K[T]$ is the minimal polynomial of α , let $h_\alpha \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

The marked-root decomposition identifies the algebra remaining after the diagonal factor with

$$D_\alpha := L[T]/(h_\alpha),$$

a finite L -algebra of degree two. Let N/K be the normal closure of L/K , containing L through the marked inclusion. Since L/K is finite separable, N/K is Galois [The26, Tag 0EXM]. Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Restriction of automorphisms gives a bijection

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \longmapsto g|_L.$$

Because the conjugates of L generate its normal closure N , an element of G fixing every conjugate embedding fixes N . Hence the coset action is faithful and

$$\text{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

The residual collision algebra detects which cubic branch occurs. We include the short argument in order to fix its marked identification with the normal closure.

Proposition 3.1 (Connected cubic residual factor). *The following conditions are equivalent:*

- (i) $\text{Spec } D_\alpha$ is connected;

- (ii) D_α is a field;
- (iii) h_α is irreducible over L ;
- (iv) L/K is nonnormal.

When they hold, there is an L -algebra isomorphism

$$\varphi: D_\alpha \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

Here $\text{pr}_1: L \times N \rightarrow L$, $(\ell, n) \mapsto \ell$, is projection onto the first factor.

Proof. Write

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$

The quadratic h_α is separable, so D_α is finite étale over L ; finite étale algebras over a field are finite products of finite separable field extensions [The26, Tag 00U3]. If h_α is irreducible, then D_α is a field and hence has connected spectrum. If it is reducible, it is a product of two distinct linear factors; the Chinese remainder theorem then gives $D_\alpha \simeq L \times L$, whose spectrum is disconnected. This proves the equivalence of (i), (ii), and (iii).

If h_α is reducible, it has a root $\beta \in L$. Separability gives $\beta \neq \alpha$, and the third root is

$$\gamma = -a_2 - \alpha - \beta \in L.$$

Thus m_α would split over L . Since $L = K(\alpha)$, the standard generator criterion for normality [The26, Tag 0BR3] shows that L/K would be normal. Conversely, if L/K is normal, then m_α splits over L , so h_α is reducible. Hence (iii) and (iv) are equivalent.

Assume these equivalent conditions, and let β denote the image of T in D_α . This field contains the three roots

$$\alpha, \quad \beta, \quad -a_2 - \alpha - \beta$$

of m_α , and is generated over L by β . It is therefore the splitting field of m_α , which is the normal closure N of L/K [The26, Tag 09DT]. Choose the resulting L -algebra isomorphism

$$\varphi: D_\alpha \xrightarrow{\sim}_L N.$$

Let

$$\Phi_\alpha: L \otimes_K L \xrightarrow{\sim}_L L \times D_\alpha$$

be the marked-root decomposition of Theorem 2.6. It satisfies

$$\text{pr}_1 \circ \Phi_\alpha = \mu_L.$$

Composing its second factor with φ , let

$$\Psi = (\text{id}_L \times \varphi) \circ \Phi_\alpha.$$

Since $\text{id}_L \times \varphi$ leaves the first factor unchanged, $\text{pr}_1 \circ \Psi = \mu_L$. □

Corollary 3.2 (The generic-degree-three counterexample locus). *The map F is not a polynomial automorphism. Consequently,*

$$F \text{ is a counterexample to } JC(3) \iff F \text{ is Keller.}$$

When these equivalent conditions hold, L/K is nonnormal and $\text{Spec } D_\alpha$ is connected.

Proof. If F were a polynomial automorphism, its coordinate functions would generate A , so $B = A$ and $L = K$, contrary to $[L : K] = 3$. The displayed equivalence now follows from the definition of a counterexample to $JC(3)$.

Assume these conditions, so that F is Keller. If L/K were normal, Theorem 2.8 would give $L = K$, again contradicting $[L : K] = 3$. Proposition 3.1 now identifies nonnormality with connectedness of $\text{Spec } D_\alpha$. $\square$

Theorem 3.3 (Galois group of the connected residual factor). *If $\text{Spec } D_\alpha$ is connected, then*

$$\text{Gal}(N/K) \cong S_3.$$

Proof. Proposition 3.1 identifies N with the degree-two field D_α over L . Hence

$$|H| = [N : L] = 2.$$

Here we use the fixed-field degree formula from the fundamental theorem of Galois theory [The26, Tag 09DW]. The normal-closure identities recalled above give

$$\text{core}_G(H) = 1, \quad [G : H] = [L : K] = 3.$$

Hence the action of G on the three cosets of H gives

$$\rho: G \longrightarrow \text{Perm}(G/H) \cong S_3, \quad \ker \rho = \text{core}_G(H) = 1.$$

Core-freeness therefore makes ρ injective, so

$$|G| \leq |S_3| = 6.$$

The index formula gives

$$|G| = [G : H]|H| = 3 \cdot 2 = 6.$$

Thus $|G| = 6$, and the injective homomorphism ρ is an isomorphism $G \cong S_3$. $\square$

3.2 The cubic residual factor

Retain the cubic data of Section 3.1 and suppose that F lies in the counterexample locus of Corollary 3.2. Equivalently, F is Keller; the same corollary shows that D_α is connected and L/K is nonnormal. Proposition 3.1 then identifies D_α with N . Thus

$$B = \mathbb{C}[F_1, F_2, F_3] \subseteq A = \mathbb{C}[x_1, x_2, x_3], \quad K = \text{Frac}(B) \subseteq L = \text{Frac}(A) = K(\alpha) \subseteq N,$$

where $[L : K] = 3$ and N/K is the normal closure. Let

$$\theta: K \otimes_B A \xrightarrow{\sim} L, \quad \lambda \otimes a \longmapsto \lambda a.$$

Corollary 3.4 (Generic cubic S_3 collision). *There is a K -algebra isomorphism*

$$\Psi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \times N, \quad \text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

It satisfies

$$\Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N, \quad \text{Gal}(N/K) \cong S_3.$$

In particular,

$$\text{Obs}(F) \neq 0, \quad I_R \subsetneq I_\Delta.$$

Proof. Theorem 2.5 gives a K -algebra isomorphism

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

such that

$$\mu_L \circ \Phi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

Proposition 3.1 gives

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

After restricting Ψ from L to K , let

$$\Psi_F = \Psi \circ \Phi_F.$$

It follows that

$$\text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F), \quad \Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N \neq 0.$$

Suppose that the diagonal-evaluation homomorphism $\bar{\mu}_F: C_F \rightarrow A$ were injective. Since K is a localization of B , it is flat over B , so $1 \otimes \bar{\mu}_F$ would also be injective. Theorem 2.5 shows that θ is an isomorphism. Therefore $\theta \circ (1 \otimes \bar{\mu}_F)$ would be injective, contradicting the nonzero kernel $0 \times N$ above. Hence $\ker(\bar{\mu}_F) \neq 0$, and Proposition 2.2 gives $\text{Obs}(F) \neq 0$. Proposition 2.1 shows that $I_R \subseteq I_\Delta$ is strict.

Finally, Theorem 3.3 identifies $\text{Gal}(N/K)$ with S_3 . □

Remark (Ordered roots). Let

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0$$

have roots $\alpha_1, \alpha_2, \alpha_3$ in N . For distinct i, j, k ,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \quad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

Thus the residual factor, which marks two distinct roots, is already the ordered-root normal closure.

Lean correspondence. `CollisionIdeals/ComplexThree/Cubic/FunctionField.lean`, `CollisionIdeals/ComplexThree/Cubic/GaloisGroup.lean`, `CollisionIdeals/ComplexThree/Cubic/S3Collision.lean`, `CollisionIdeals/ComplexThree/Cubic/Branch.lean`, `CollisionIdeals/ComplexThree/Cubic/Main.lean`, and `CollisionIdeals/ComplexThree/Statements/JacobianConjecture.lean`. The descent from a nontrivial generic residual factor to $\text{Obs}(F) \neq 0$ is formalized in `CollisionIdeals/General/GenericFiber/DiagonalObstruction.lean`. The marked normal-closure object and fixing subgroup are in `CollisionIdeals/General/Galois/NormalClosure.lean`. The generic Lean cubic witness builder takes the residual-field isomorphism and $H \neq 1$ as explicit inputs. The cubic branch module now derives

both from nonnormality: the dimension-independent normal-closure API gives $H \neq 1$, and the cubic normal-closure/Galois-action and dimension-count argument constructs the marked residual-field isomorphism. The dimension-three wrapper uses these two implications. It proves obstruction nonvanishing before using the Keller hypothesis solely to assemble the counterexample statement; the exact generic-degree-three equivalence displayed above is not yet packaged as a separate Lean theorem. The full connectedness, field, irreducibility, and nonnormality equivalence is likewise not yet packaged: the branch module proves the nonnormal-to-irreducible direction and the residual normal-closure identification. In the manuscript the two field-theoretic inputs follow from Proposition 3.1 and Corollary 3.2. Classical Galois rigidity is represented by the theorem-target proposition `ComplexKellerGaloisRigidity`, but is neither assumed as an axiom nor proved in Lean. Hidden inertia is formalized separately, pointwise at a supplied ramified divisor and conditional on a supplied Keller collision-normalization model and $H \neq 1$, in `CollisionIdeals/ComplexThree/Cubic/HiddenInertia.lean`. Thus the cubic core has no scheme-normalization or ramification dependency.

Remark (Exact relation with classical complex $JC(3)$). The traditional complex three-dimensional Jacobian conjecture asserts that every polynomial map

$$F: \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3, \quad \det JF \in \mathbb{C}^\times,$$

is a polynomial automorphism; equivalently, every such map satisfies $\text{Obs}(F) = 0$. In generic degree three, Corollary 3.2 instead gives

$$\begin{aligned} [L : K] = 3 &\implies \text{Obs}(F) \neq 0, & F \notin \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3), \\ F \text{ is a counterexample to } JC(3) &\iff \det JF \in \mathbb{C}^\times, \\ F \text{ is a counterexample} &\implies \text{Spec } D_\alpha \text{ connected,} & \text{Gal}(N/K) \cong S_3. \end{aligned}$$

The first line is independent of the Keller condition. The additional content on the counterexample locus is that Theorem 2.8 excludes the normal cubic branch, after which the residual algebra itself produces the normal closure and its Galois group S_3 . Thus $JC(3)$ implies that the generic-degree-three counterexample locus is empty. This section does not construct such a map, and it does not assert that every possible counterexample to $JC(3)$ has generic degree three.

4 Further directions

The normal closure also carries a dimension-independent boundary picture, although none of it is needed for the cubic collision theorem. Let L/K be finite separable, let N/K be its normal closure, and let Y be a normal integral model with function field K . Write

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L), \quad Z = \text{Norm}_N(Y) \longrightarrow \overline{X} = \text{Norm}_L(Y) \longrightarrow Y.$$

Suppose that a normal model X of L is an open subscheme of $\overline{X}$ over Y , and that $X \rightarrow Y$ is étale.

Proposition 4.1 (Hidden inertia on a conjugate sheet). *Let $E \in Z^{(1)}$ be ramified over Y , with divisorial valuation w_E , restriction $v = w_E|_K$, and inertia group I_E . Then some $g \in G$ satisfies*

$$e((g^{-1}w_E)|_L/v) = [I_E : I_E \cap gHg^{-1}] > 1.$$

For every such g , the center of $(g^{-1}w_E)|_L$ on $\overline{X}$ lies in $\overline{X} \setminus X$.

Proof. Because N is the normal closure of L/K , the action of G on G/H is faithful, so $\bigcap_{g \in G} gHg^{-1} = 1$. Since $I_E \neq 1$, it is not contained in every conjugate of H ; the usual inertia formula gives the displayed index for a suitable g . If the corresponding center belonged to X , the étale morphism $X \rightarrow Y$ would force its ramification index to be one, a contradiction. $\square$

On the generic-degree-three counterexample locus of Section 3, the cubic extension is necessarily nonnormal and $G \cong S_3$. Thus any divisorial ramification in the ordered-root normalization is detected on a conjugate sheet and forced into the deleted boundary. This statement is conditional on the existence of such a ramified divisor; the cubic collision theorem itself neither requires nor asserts it.

The cubic and planar problems use the same diagonal in opposite ways. Here generic degree three forces a nonzero residual factor beside it. In the planar sequel, the goal is to prove that the off-diagonal factor vanishes, so that the collision scheme consists only of the diagonal.

A sequel will develop this normalization picture for planar Keller maps: conjugate affine sheets, their boundary and fixed-locus ideals, and the secant–trace comparison. Its remaining Keller-specific input is a uniform landing theorem converting a finite secant–frame lattice into control of arbitrary boundary sections. Purity and finite-étale rigidity enter only after that boundary statement is established.

5 Conclusion

The collision ideal turns injectivity into a scheme-theoretic equality. The obstruction $\text{Obs}(F)$ is precisely the kernel of diagonal evaluation, and it vanishes exactly when the collision relation equals the diagonal. After passage to the generic fiber, that diagonal becomes the multiplication factor of $L \otimes_K L$, while the remaining factors record the other embeddings of L .

In generic degree three, the diagonal persists as the first factor beside a quadratic residual algebra, and nonautomorphy is automatic. Thus the counterexample condition is equivalent to the Keller condition. On this locus the classical Galois case rules out the split, normal branch; hence the residual algebra is connected, it is the normal closure N , its Galois group is S_3 , and

$$K \otimes_B C_F \simeq L \times N.$$

Its nonzero residual factor gives $\text{Obs}(F) \neq 0$. Hence every map in the generic-degree-three counterexample locus has the normal-closure factor N , with $\text{Gal}(N/K) \cong S_3$. The theorem does not assert that this locus is nonempty or classify counterexamples of other generic degrees. The normalization principle of Section 4 explains how the same conjugate sheet geometry leads to boundary questions, which are reserved for the planar sequel.

Motivating example. The explicit three-dimensional polynomial counterexample announced by Levent Alpöge [Alp26] and independently formalized in Isabelle/HOL by Ramos, Hulak, and de Queiroz [RHdQ26] motivated the cubic S_3 comparison. The formal verification establishes a constant nonzero Jacobian determinant and distinct rational points with the same image. That particular map is not used in the arguments above: the dimension-three result describes the structure of the generic-degree-three counterexample locus.

The formalization uses Lean 4 and Mathlib [dMU21, The20].

Corollary 5.1 (First possible generic degree). *If F is Keller and $[L : K] \leq 2$, then F is a polynomial automorphism. Consequently, every complex Jacobian-conjecture counterexample has generic degree at least three.*

Proof. Proposition 2.4 makes L/K finite and separable. In degree one it is trivial, and in degree two it is Galois. Theorem 2.8 applies in either case. $\square$

This low-degree boundary is the handoff to the planar sequel. Paper II asks for the stronger conclusion that every planar Keller map has generic degree one—equivalently, by Galois rigidity and the automorphism criterion, that its collision scheme collapses to the diagonal.

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